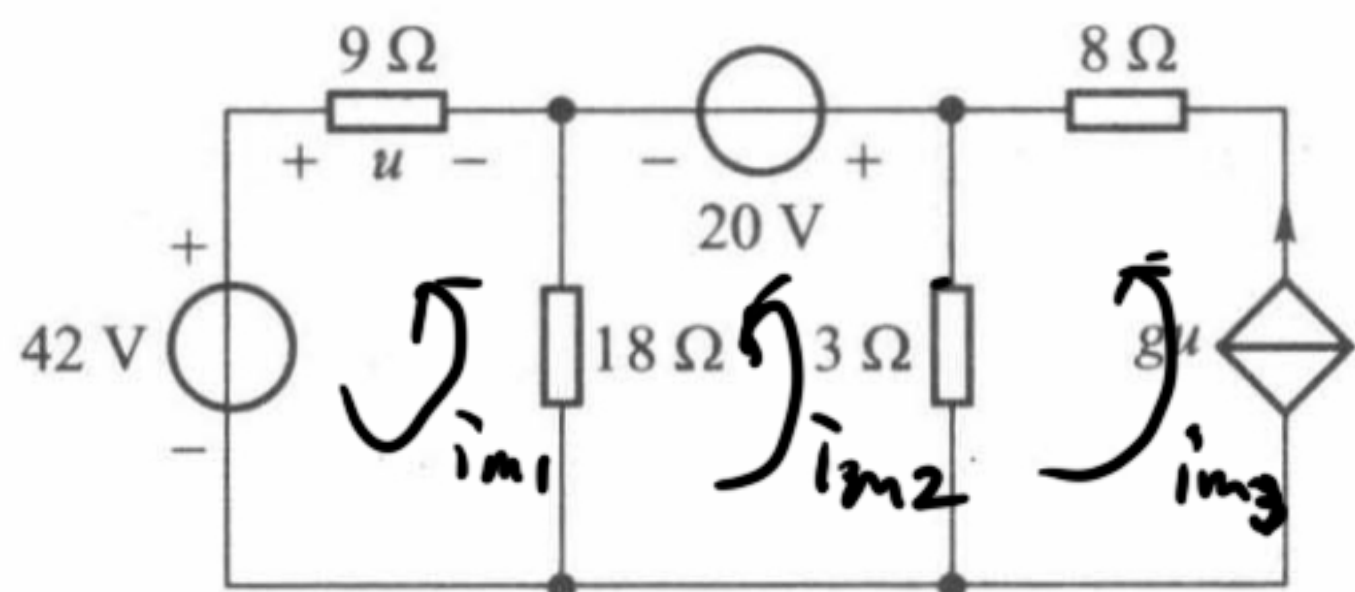


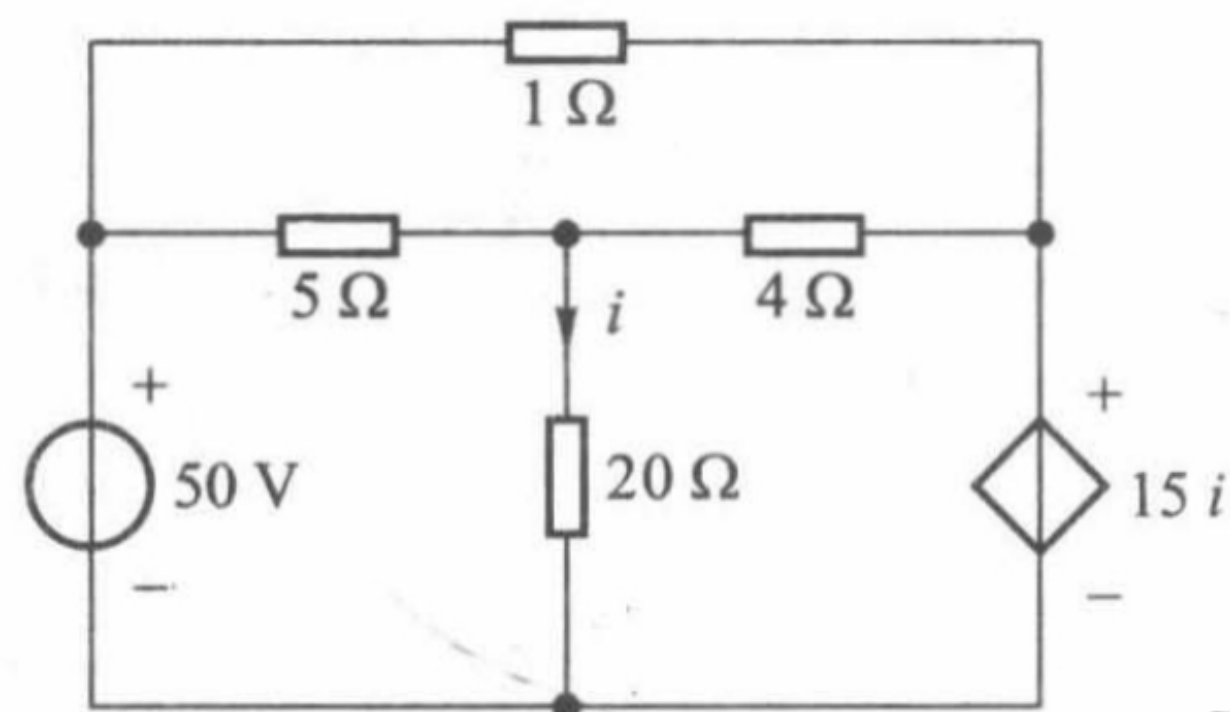
第二章习题

2-9 电路如图题 2-7 所示, 其中 $g=0.1\text{ S}$, 用网孔分析求流过 $8\ \Omega$ 电阻的电流。

2-10 含 CVCVS 电路如图题 2-8 所示, 试求受控源功率。



图题 2-7



图题 2-8

$$i_{m3} = -0.9 i_{m1}$$

$$9 i_{m1} - 18 i_{m2} + 42 = 0$$

$$16.2 i_{m1} = -69$$

$$\frac{9 \times 7}{2 \times 8} \times 7 \quad 4.5 \times 7 = \frac{80}{31.5}$$

$$27 i_{m1} - 18 i_{m2} = -42$$

$$31.5 i_{m1} - 21 i_{m1} = -49$$

$$4 i_{m2} - 15.3 i_{m1} = -20$$

$$\begin{cases} i_{m1} \cdot 18 + i_{m1} \cdot 9 + 42 - 18 i_{m2} = 0 \\ 3(i_{m2} - i_{m3}) + 20 + 18(i_{m2} - i_{m1}) = 0 \\ i_{m3} = g \cdot u \\ g = -\frac{u}{i_{m1}} \end{cases}$$

$$\Rightarrow \begin{cases} 27 i_{m1} - 18 i_{m2} = -42 \\ 21 i_{m2} - 15.3 i_{m1} = -20 \\ i_{m3} = -0.9 i_{m1} \end{cases}$$

$$\Rightarrow \begin{cases} i_{m1} = -4.26\text{ A} \\ i_{m2} = -4.06\text{ A} \\ i_{m3} = 3.83\text{ A} \end{cases}$$

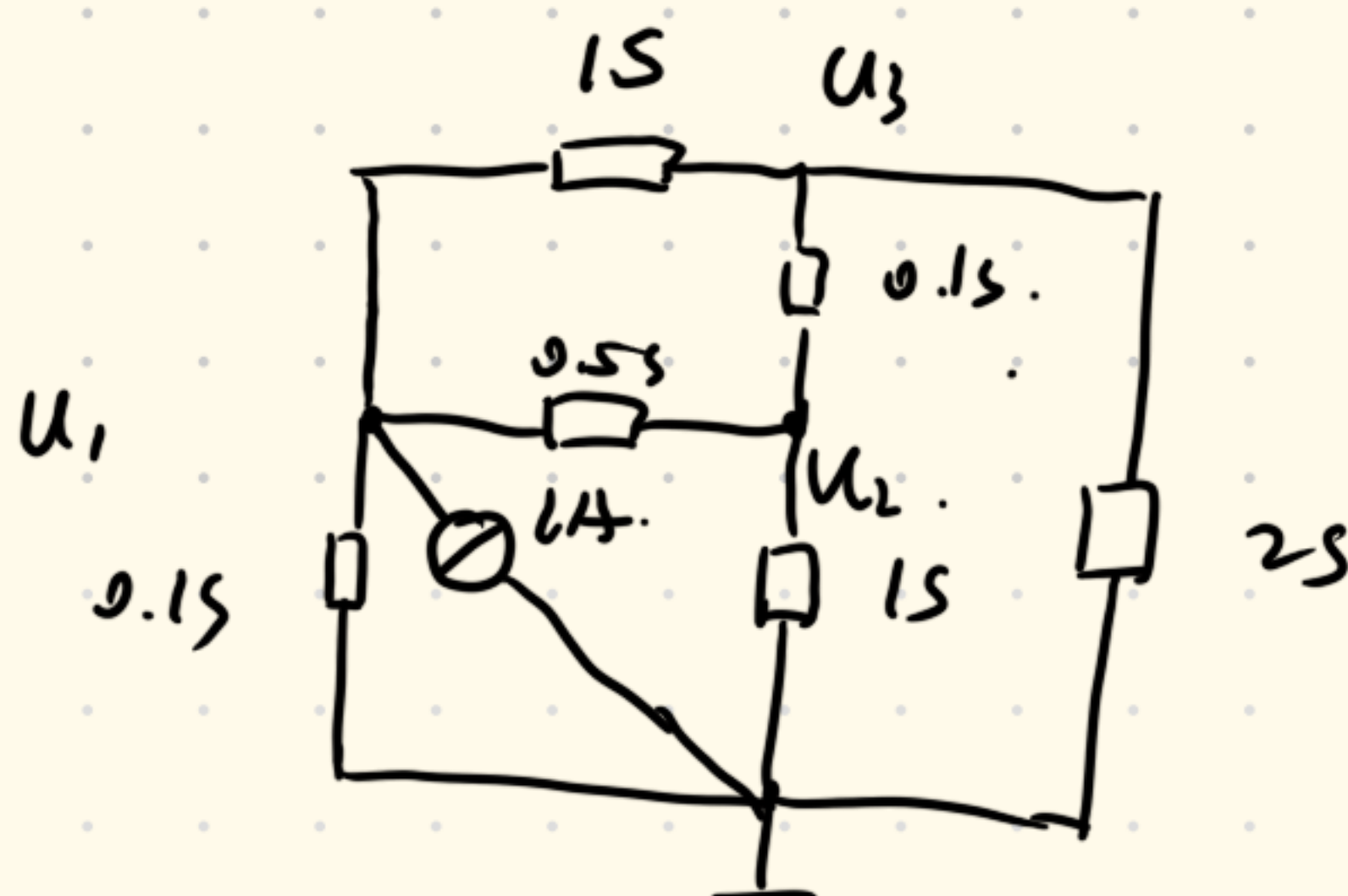
∴ 通过 $8\ \Omega$ 电阻的电流为 3.83 A , 从右向左为正方向。

2-15 若节点方程为

$$1.6u_1 - 0.5u_2 - u_3 = 1$$

$$-0.5u_1 + 1.6u_2 - 0.1u_3 = 0$$

$$-u_1 - 0.1u_2 + 3.1u_3 = 0$$



$$\begin{cases} G_{11}u_{N1} + G_{12}u_{N2} + G_{13}u_{N3} = i_{S11} \\ G_{21}u_{N1} + G_{22}u_{N2} + G_{23}u_{N3} = i_{S22} \\ G_{31}u_{N1} + G_{32}u_{N2} + G_{33}u_{N3} = i_{S33} \end{cases}$$

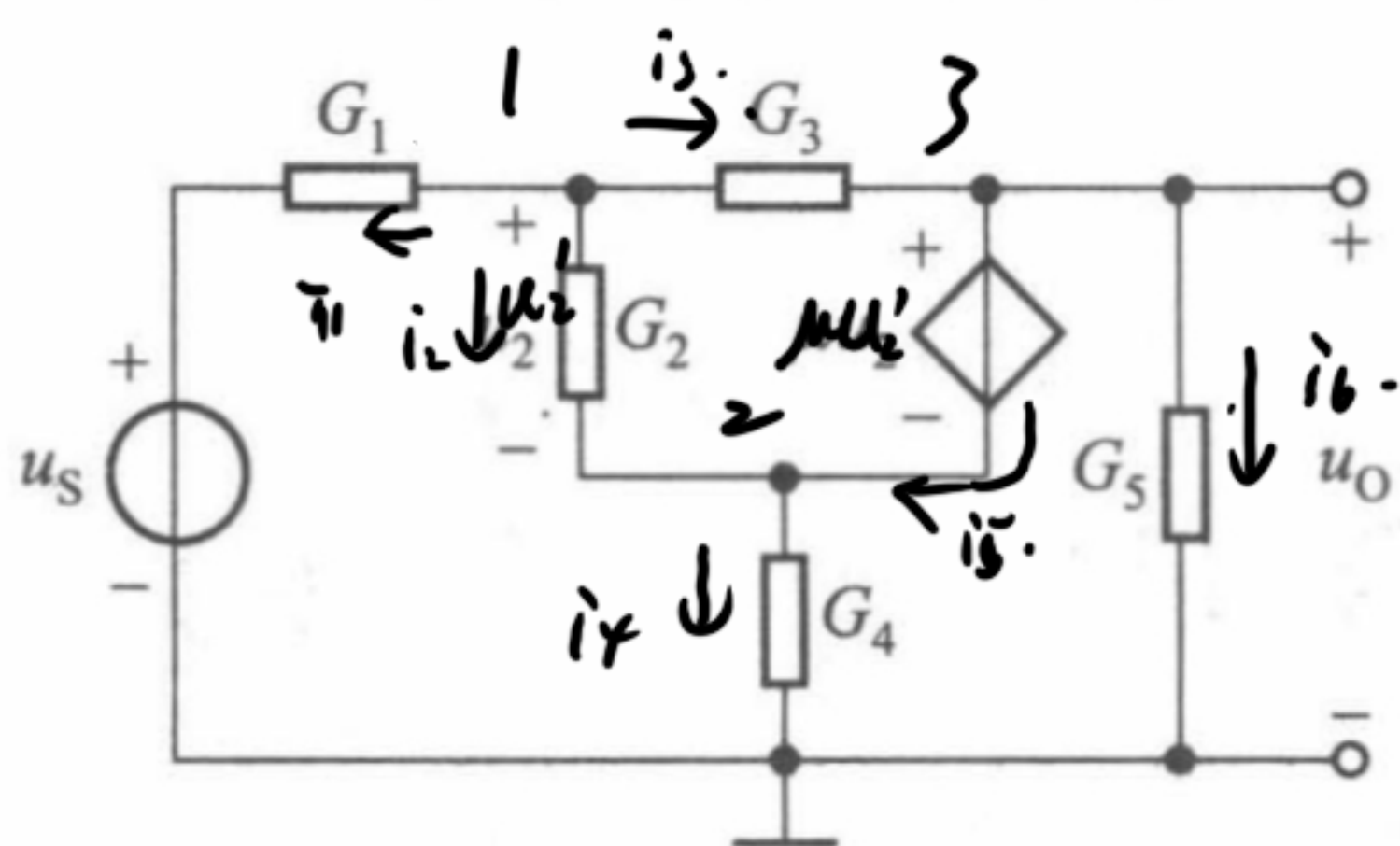
如下处理

2-20 试列出为求解图题 2-17 所示电路中 u_0 所需的节点方程。

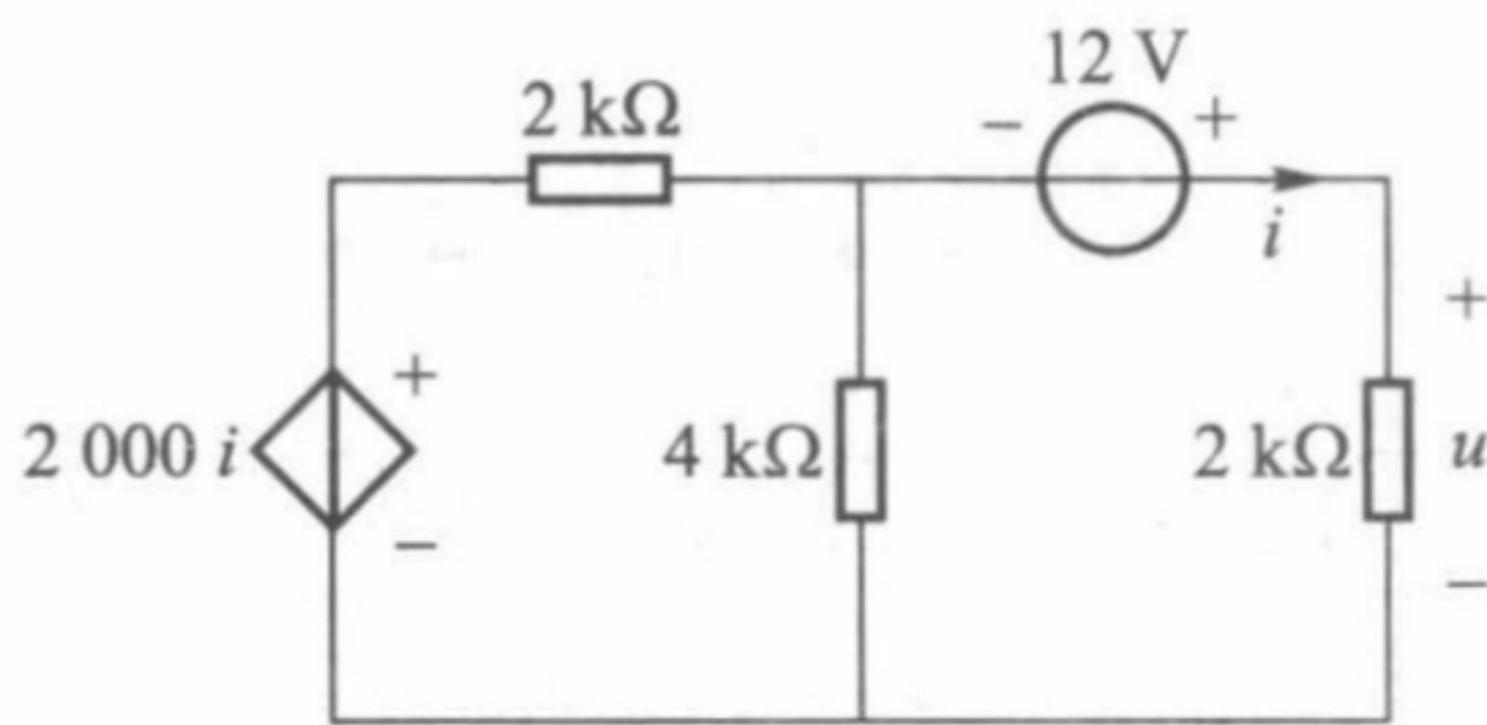
2-21 试求图题 2-18 所示电路电压 u 。

第一篇 总论和电阻电路的分析

$$i_1 - \frac{1}{G_1} + u_s = u_1$$



图题 2-17



图题 2-18

节点 1: $i_1 + i_2 + i_3 = 0$

即为 $G_1(u_1 - u_s) + G_2(u_2 - u_1) + G_3(u_3 - u_1) = 0$

整理得

节点 2: $i_2 + i_5 = i_4$

即: $G_2(u_2 - u_1) + i_5 = G_4 u_2$

节点 3: $i_3 = i_5 + i_6$

$G_3(u_3 - u_1) = i_5 + G_5 u_3$

补充方程: $u_1 - u_2 = u_2'$
 $u_3 - u_2 = \mu u_2'$
 $u_0 = u_3$

$$\begin{cases} (G_1 - G_2 - G_3)u_1 + G_2 u_2 + (G_3 - G_1)u_3 = 0 \\ (G_2 - G_4)u_2 + (G_3 - G_5)u_3 - (G_1 + G_3)u_1 = 0 \\ (u_3 - u_2) = \mu(u_1 - u_2) \\ u_0 = u_3 \end{cases}$$

4个未知量 4个方程

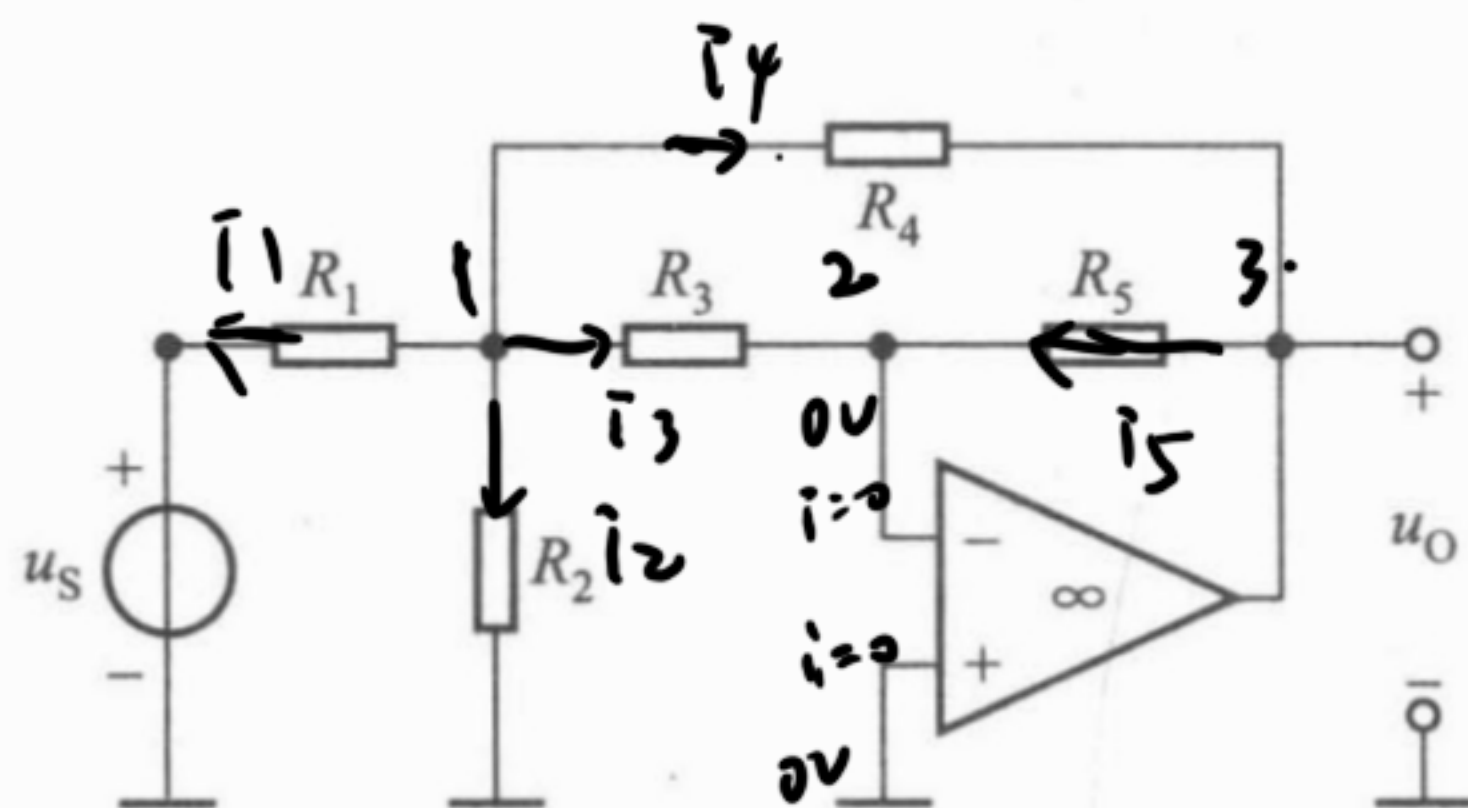
2-22 图题 2-19 所示运放电路, $R_1=R_2, R_4=R_5$, 试表明

$$u_o = -\frac{R}{R_1} u_s$$

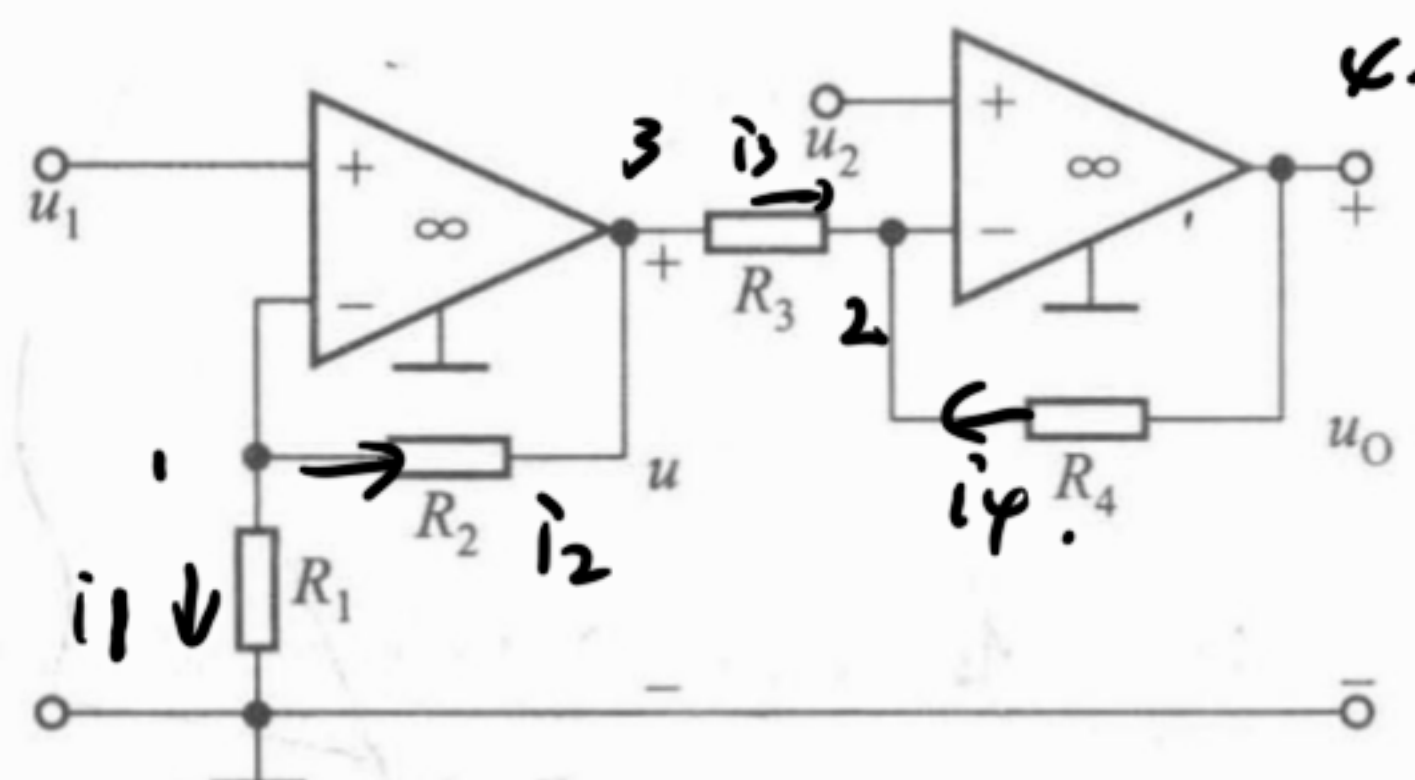
式中

$$\frac{1}{R} = \frac{2R_3}{R_1 R_4} + \frac{2}{R_4} + \frac{R_3}{R_4^2}$$

2-23 图题 2-20 所示运放电路, 试证明 $u_o = \left(1 + \frac{R_4}{R_3}\right) u_2 - \frac{R_4}{R_3} \left(1 + \frac{R_2}{R_1}\right) u_1$



图题 2-19



图题 2-20

2-22: 节点 1: $i_1 + i_2 + i_3 + i_4 = 0$

$$\frac{u_1 - u_s}{R_1} + \frac{u_1}{R_2} + \frac{u_1}{R_3} + \frac{u_1 - u_3}{R_4} = 0 \Leftrightarrow \frac{2u_1 - u_s}{R_1} + \frac{u_1}{R_3} + \frac{u_1 - u_3}{R_4} = 0 \quad \text{--- (1)}$$

节点 2: $i_3 + i_5 = 0$

$$\frac{u_1}{R_3} + \frac{u_3}{R_5} = 0 \Leftrightarrow \frac{u_1}{R_3} = -\frac{u_3}{R_4} \quad \text{--- (2)}$$

$$u_3 = u_o$$

$$\Leftrightarrow u_1 = u_o \quad \text{--- (3)}$$

①③联立

$$u_1 = -\frac{R_3}{R_4} u_o \quad \text{--- (5)}$$

③⑤代入 ① 得

$$\frac{2(-\frac{R_3}{R_4} u_o) - u_s}{R_1} + \frac{-\frac{R_3}{R_4} u_o}{R_3} + \frac{-\frac{R_3}{R_4} u_o - u_o}{R_4} = 0$$

$$\Leftrightarrow -\frac{2R_3}{R_1 R_4} u_o - \frac{1}{R_1} u_s - \frac{1}{R_4} u_o - \frac{R_3}{R_4^2} u_o - \frac{1}{R_4} u_o = 0$$

$$\Leftrightarrow -\left(\frac{2R_3}{R_1 R_4} + \frac{2}{R_4} + \frac{R_3}{R_4^2}\right) u_o = \frac{1}{R_1} u_s \quad \text{--- (6)}$$

$$\text{根据题意得} \quad -\frac{u_o}{R} = \frac{1}{R_1} u_s \quad \text{--- (7)}$$

$$\therefore u_o = -\frac{R}{R_1} u_s$$

∴ 得证!

2-23:

节点1: $i_1 + i_2 = 0$

即 $\frac{U_1}{R_1} + \frac{U_1 - U_3}{R_2} = 0 \Rightarrow (1 + \frac{R_2}{R_1}) U_1 = U_3$

节点2: $i_3 + i_4 = 0$

即: $\frac{U_0 - U_2}{R_4} + \frac{U_3 - U_2}{R_3} = 0$

$\therefore U_0 = U_2 + \frac{R_4}{R_3} (U_2 - U_3)$

代入.

$\therefore U_0 = (1 + \frac{R_4}{R_3}) U_2 - \frac{R_4}{R_3} \cdot (1 + \frac{R_2}{R_1}) \cdot U_1$

$\therefore$ 得证!